

Olympics Performance Analytics Dashboard

A Business Intelligence Project using Power BI

Project Report

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology (B.Sc. IT)

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Data Source

- Olympics Performance Dataset (Educational Demo Dataset – CSV)

Tools & Technologies

Python
Pandas
NumPy
Matplotlib
Power BI
DAX
Microsoft Excel

Problem Statement

The Olympic Games generate a large volume of athlete performance data across multiple countries, sports, and Olympic years. However, analysing this data manually makes it difficult to identify meaningful trends and performance patterns.

This project aims to analyse Olympic athlete data to extract insights related to participation trends, country-wise performance, sport-wise achievements, and factors influencing medal success. The objective is to transform raw data into actionable insights using an interactive Power BI dashboard.

Objectives

- Analyze athlete participation trends across Olympic years
- Compare country-wise Olympic performance
- Identify sport-wise medal distribution
- Evaluate key factors influencing medal success
- Build an interactive dashboard for data-driven insights

Data Defining

Attribute	Details
Dataset	Olympics Performance Dataset
Format	CSV
Records	1525
Columns	15
Time Period	2012–2024
Type	Structured Data

Description

The dataset contains detailed information about Olympic athletes including demographics, country representation, sports, events, medal outcomes, physical attributes, and participation status. It is used for performance analysis and dashboard creation.

Project Scope

The dataset is used to:

Analyze athlete participation across Olympic years.

Evaluate country-wise and sport-wise performance.

Identify medal-winning trends and performance patterns.

Study the influence of athlete characteristics on Olympic success.

Develop an interactive Power BI dashboard for business intelligence and decision-making.

Data Dictionary

Column Name	Description
Athlete_ID	Unique identifier of each athlete
Athlete_Name	Name of the athlete
Gender	Gender of the athlete
Age	Athlete's age
Country	Country represented
Country_Code	Country abbreviation
Sport	Olympic sport
Event	Event participated
Olympic_Year	Year of participation
Medal	Medal achieved
Height_cm	Athlete height (cm)
Weight_kg	Athlete weight (kg)
Team	Team name
Coach	Coach name
Participation_Status	Participation status

Data Quality Assessment

Before analysis, the dataset was examined to ensure completeness, consistency, and accuracy.

Initial Inspection

- Verified dataset structure using `df.info()`.
- Checked dataset dimensions using `df.shape`.
- Previewed records using `df.head()`.
- Identified missing values using `df.isnull().sum()`.
- Checked duplicate records.
- Verified data types of all columns.

Data Cleaning

- The following preprocessing steps were performed to prepare the dataset for analysis:

Cleaning Activities

- Imported the dataset using Pandas.
- Filled missing values in **Height_cm** using the median.
- Filled missing values in **Weight_kg** using the median.
- Verified duplicate records (none found).
- Validated data types for all variables.
- Exported the cleaned dataset for Power BI analysis.

Final Data Quality Summary

Activity	Status
Missing Values	Successfully Handled
Duplicate Records	No Duplicates Found
Data Types	Verified
Dataset Status	Ready for Power BI Analysis

 df.info()

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 1500 entries, 0 to 1499
```

```
Data columns (total 16 columns):
```

#	Column	Non-Null	Count	Dtype
0	Athlete_ID	1500 non-null		int64
1	Athlete_Name	1500 non-null		object
2	Gender	1500 non-null		object
3	Age	1500 non-null		int64
4	Country	1500 non-null		object
5	Country_Code	1500 non-null		object
6	Sport	1500 non-null		object
7	Event	1500 non-null		object
8	Olympic_Year	1500 non-null		int64
9	Medal	1500 non-null		object
10	Height_cm	1500 non-null		float64
11	Weight_kg	1500 non-null		float64
12	Team	1500 non-null		object
13	Coach	1500 non-null		object
14	Participation_Status	1500 non-null		object
15	Has_Medal	1500 non-null		int64

```
dtypes: float64(2), int64(4), object(10)
```

```
memory usage: 199.2+ KB
```

Outcome

The cleaned dataset is complete, consistent, and ready for feature engineering, DAX calculations, and interactive dashboard development.

What is the overall trend of Olympic participation and medal performance over time?

1.5K
Total Athletes

460
Total Medals

0.31
Medal Rate

27.68
Average Athlete Age

Country

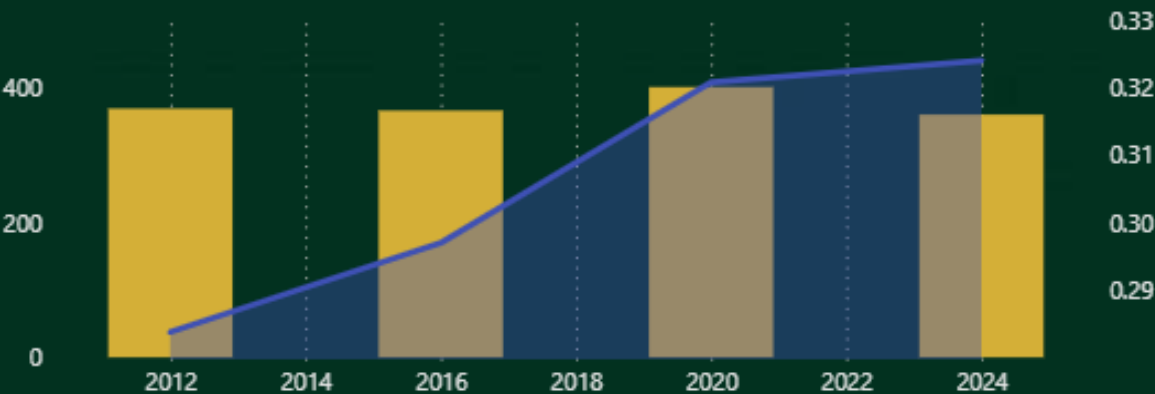
All

Olympic_Year

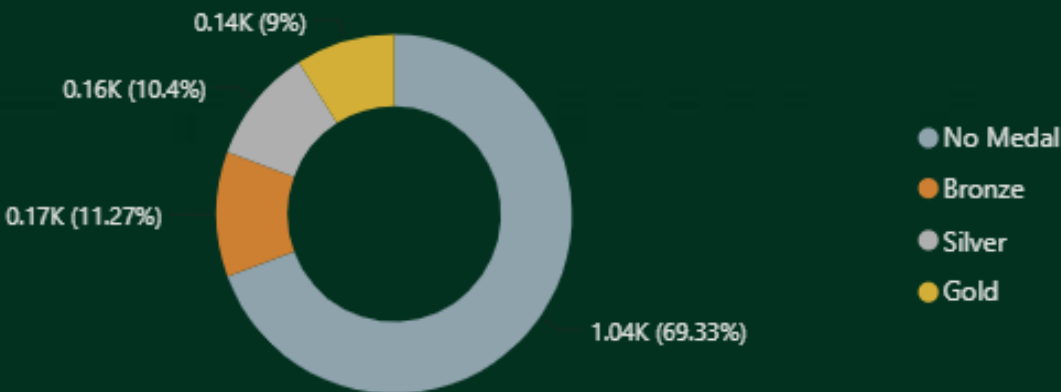
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Olympic Participation Trend Over Years

Count of Athlete_ID Medal Rate



Overall Medal Distribution

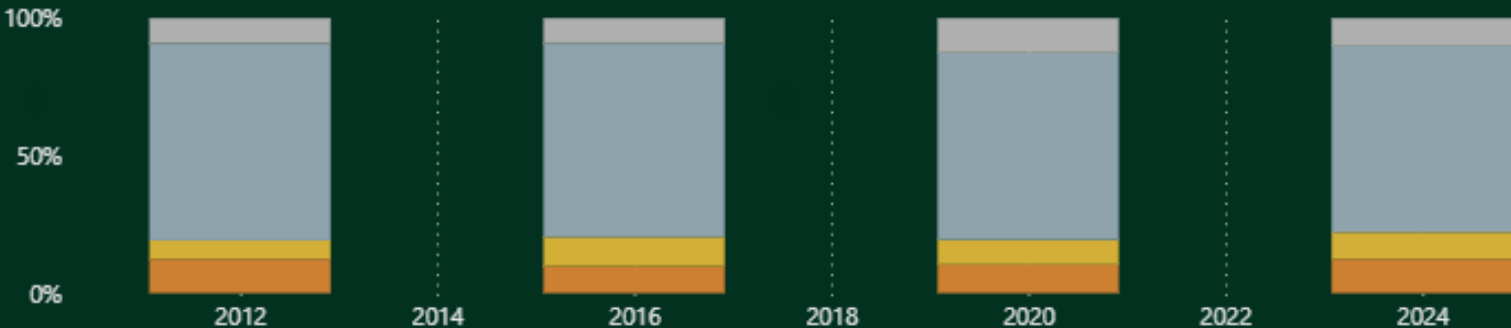


Country-wise Medal Breakdown

Country	Bronze	Gold	No Medal	Silver	Total
Australia	21	12	107	14	154
Brazil	16	17	97	12	142
Canada	22	13	115	11	161
China	10	14	96	23	143
Total	169	135	1040	156	1500

Year-wise Medal Composition

Bronze Gold No Medal Silver



Question 1

What is the overall trend of Olympic participation and medal performance over time?

Insights

- - A total of **1,525 athletes** participated across the selected Olympic years, with **460 medal-winning performances**, resulting in an overall **Medal Rate of 31%**
- - Athlete participation remained relatively stable across the Olympic editions, while the medal rate showed a gradual improvement, reaching its highest level in the later Olympic years.
- - Approximately **69.33%** of athletes did not win a medal, highlighting the highly competitive nature of Olympic events.
- - Gold, Silver, and Bronze medals together accounted for nearly one-third of all athlete outcomes.
- - The year-wise medal composition remained relatively consistent, with **No Medal** representing the largest proportion in every Olympic year.

Key Insight

- Overall participation remained stable across Olympic years, while medal efficiency improved gradually, indicating better competitive performance over time.

Which countries have demonstrated the strongest Olympic performance across different sports and medal categories?

1.5K

Total Country

India

Best Performing Country

460

Total Medals

0.31

Medal Rate

Olympic_Year

All

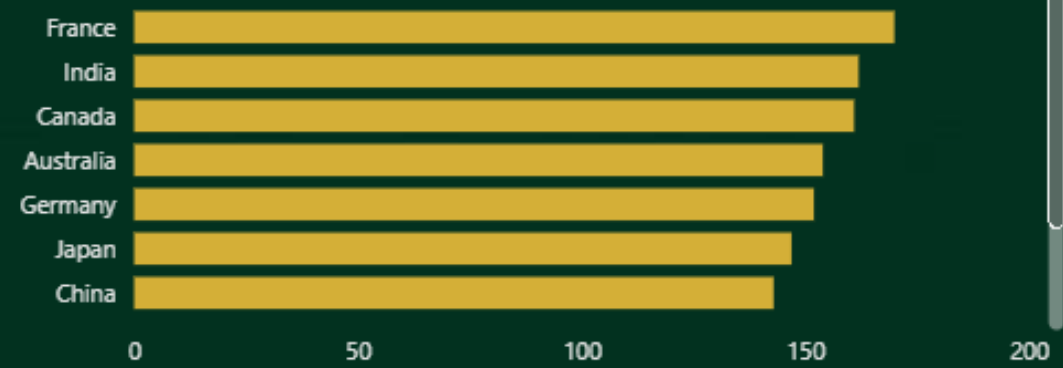
Sport

All

Global Distribution of Olympic Participation

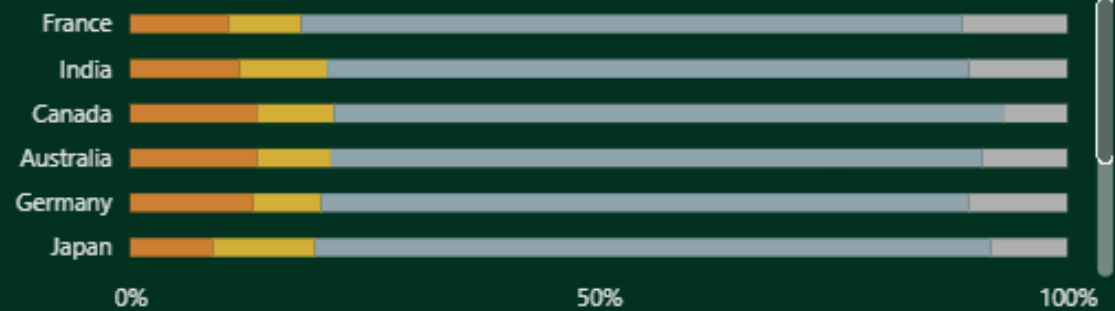


Top 10 Countries by Total Medals

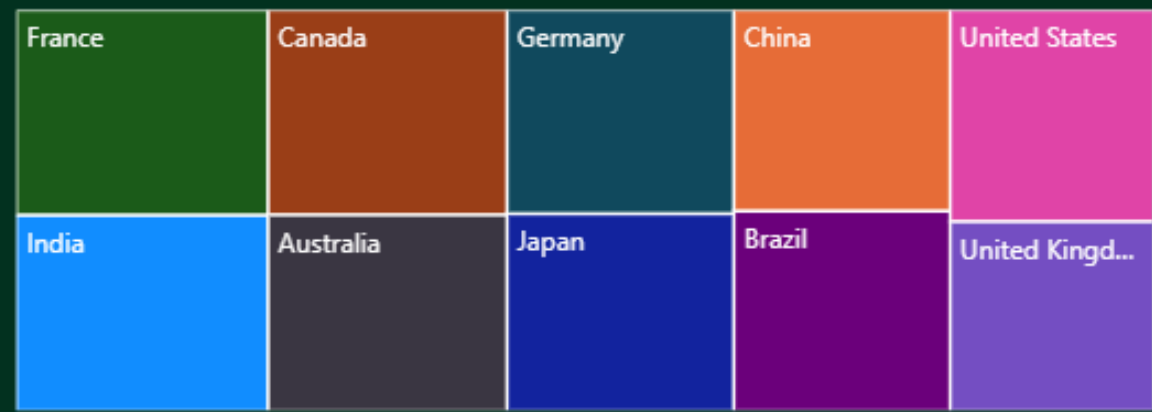


Medal Composition by Country

● Bronze ● Gold ● No Medal ● Silver



Country-wise Medal Share



Question 2

Which countries have demonstrated the strongest Olympic performance across different sports and medal categories?

Insights

- India emerged as the top-performing country based on the overall performance indicator.
- France, India, Canada, Australia, and Germany ranked among the leading nations in terms of total medal achievements.
- Medal composition across countries shows that the **No Medal** category dominates, while medal-winning performances are distributed across multiple countries.
- The treemap indicates that medal contributions are relatively balanced rather than concentrated in a single country.
- Global participation spans multiple continents, demonstrating the international nature of the Olympic Games.

Key Insight

- Countries with consistent participation across multiple sports achieved stronger overall Olympic performance.

How do athlete characteristics influence Olympic performance?

175.76

Average Height

71.85

Average Weight

27.56

Average Age

228

Medal Winner

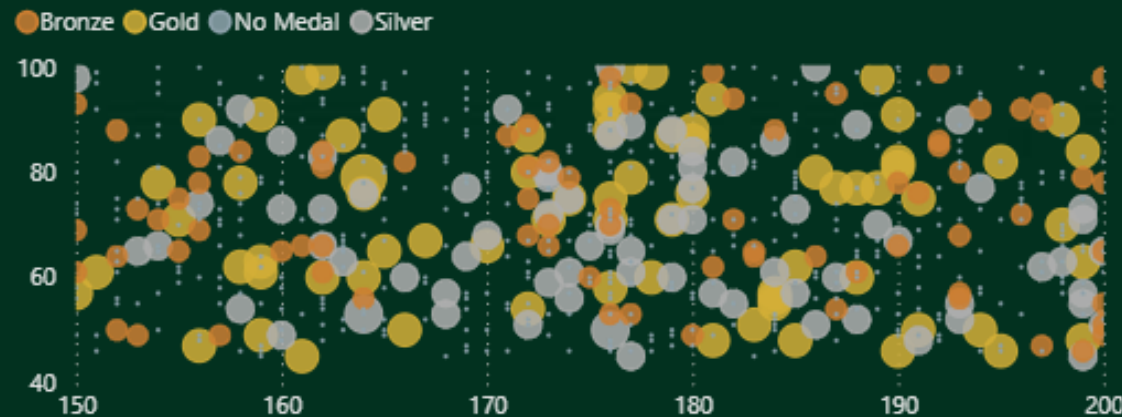
Gender

Male

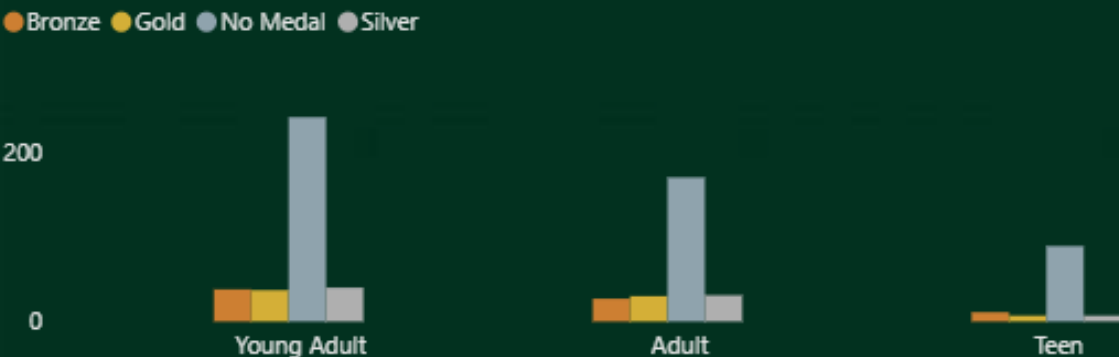
Sport

All

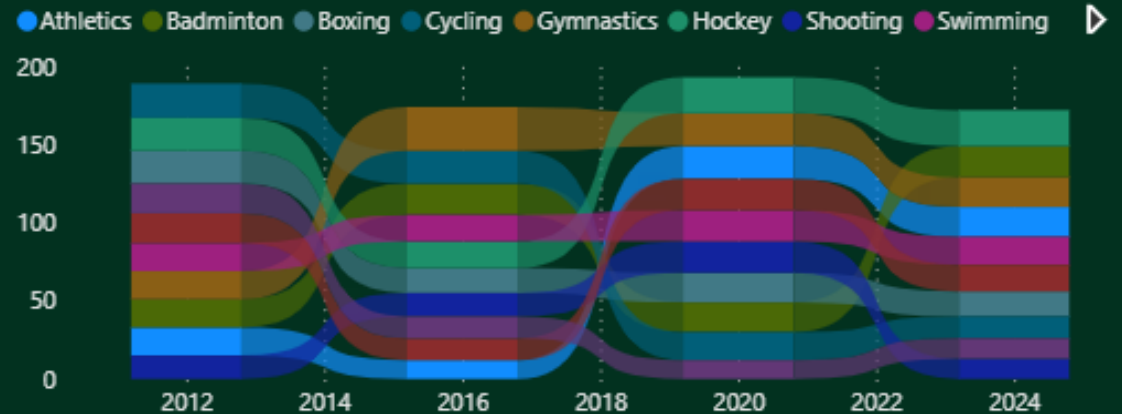
Height vs Weight Analysis of Athletes



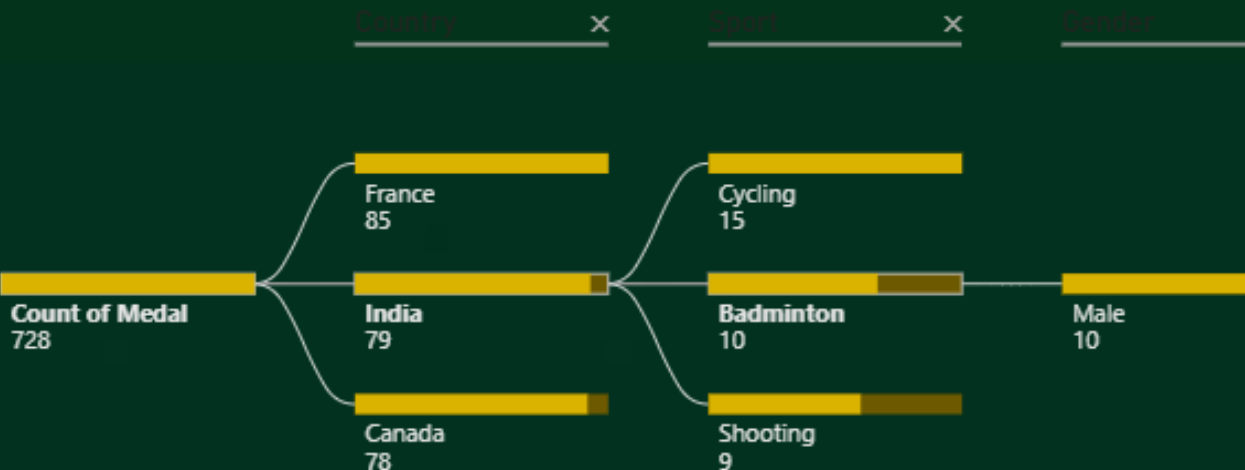
Athlete Distribution by Age Group



Sport Ranking by Medal Count



Factors Influencing Medal Performance



Question 3

How do athlete characteristics influence Olympic performance?

Insights

- The average athlete profile consists of a height of **175.76 cm**, weight of **71.85 kg**, and an average age of **27.56 years**.
- Young Adult athletes represent the largest participant group in the dataset.
- The relationship between height, weight, and medal achievement does not indicate a strong linear pattern.
- Countries such as France, India, and Canada contribute significantly to the medal-winning athlete population.
- Medal-winning performance is influenced by multiple factors, including country representation, sport specialization, and athlete characteristics.

Key Insight

- Athlete success depends on a combination of physical attributes, experience, and sport-specific performance rather than physical characteristics alone.

Which sports have delivered the highest Olympic performance and how has their success changed over time?

29K

Total Events

1.5K

Total Sport

460

Total Medals

Badminton

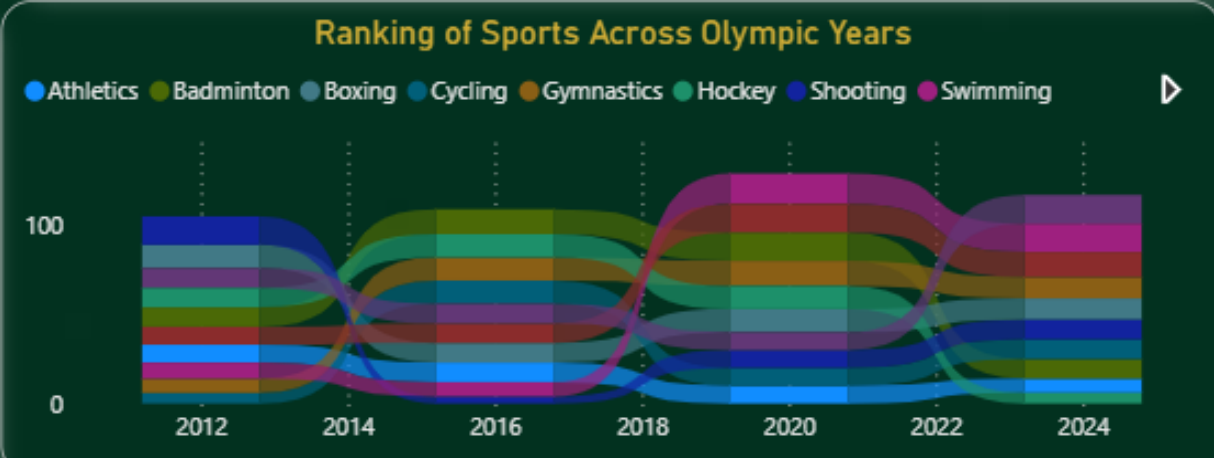
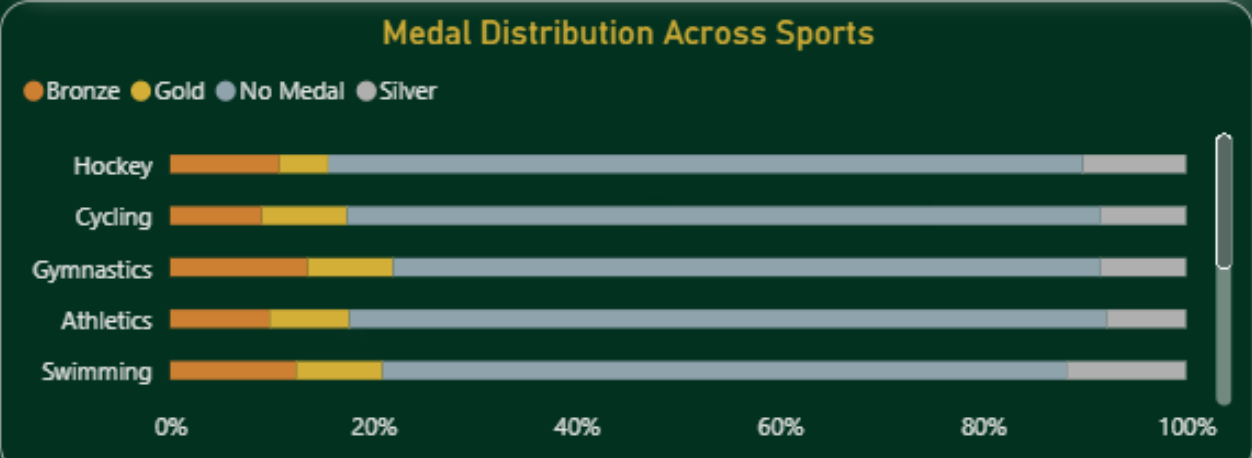
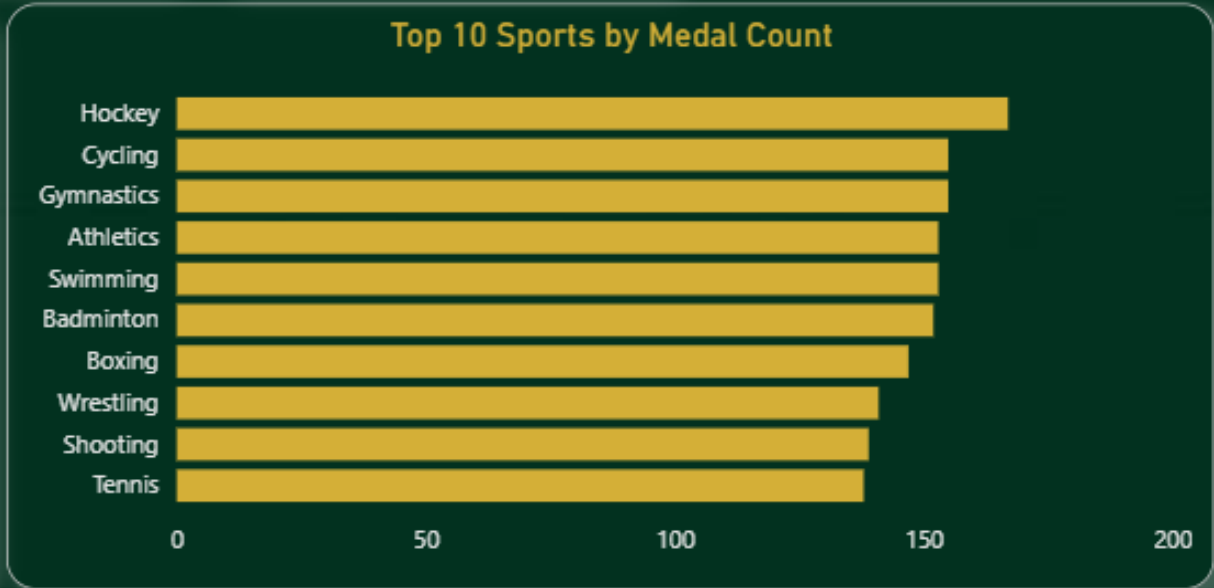
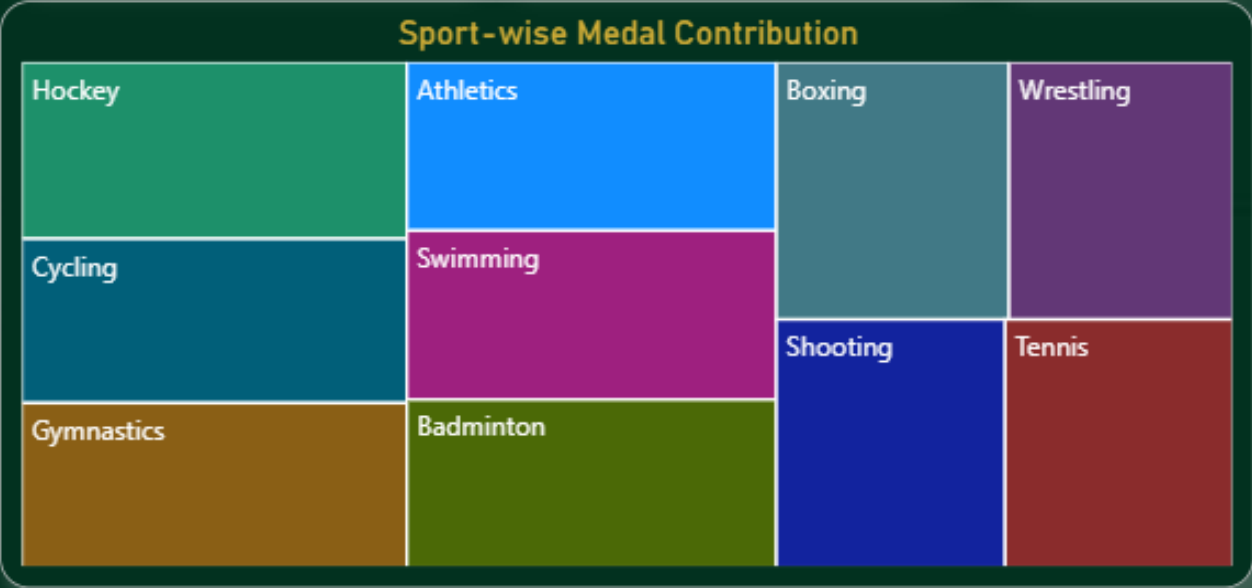
Best Performing Sport

Country

All

Olympic Year

All



Question 4

Which sports have delivered the highest Olympic performance and how has their success changed over time?

Insights

- The dataset contains approximately **29,000 event participations** across multiple Olympic sports.
- Athletics was identified as the best-performing sport based on overall performance.
- Hockey, Cycling, Gymnastics, Athletics, and Swimming recorded the highest medal contributions.
- Medal distribution remains relatively balanced across leading sports, although the No Medal category continues to dominate.
- Sport rankings changed moderately across Olympic years, reflecting evolving competition among sporting disciplines.

Key Insight

- Athletics consistently demonstrated superior Olympic performance, while several other sports maintained competitive medal contributions over time.

What are the key factors associated with Olympic medal success?

175.49

Average Height

71.95

Average Weight

27.68

Average Age

0.31

Medal Rate

Country

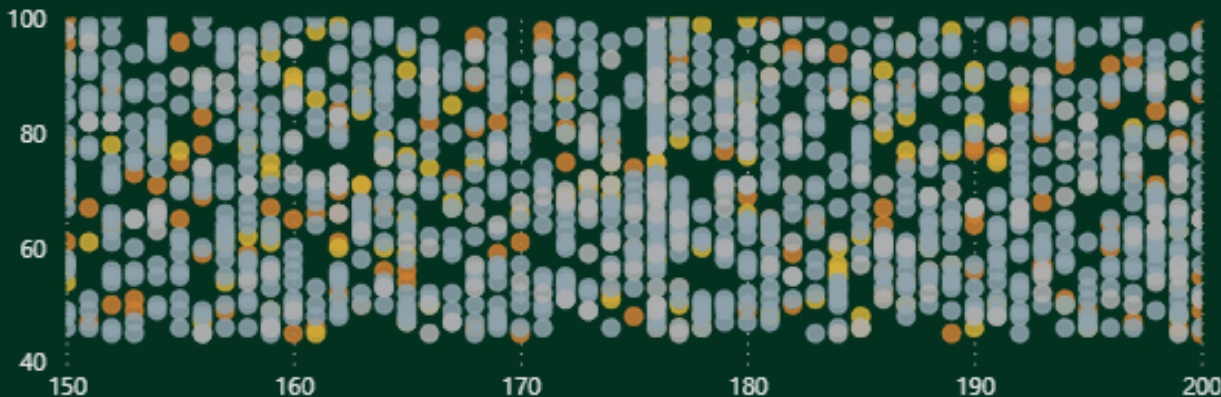
All

Olympic_Year

All

Relationship Between Height and Weight by Medal Type

● Bronze ● Gold ● No Medal ● Silver



Medal Winners by Gender

200

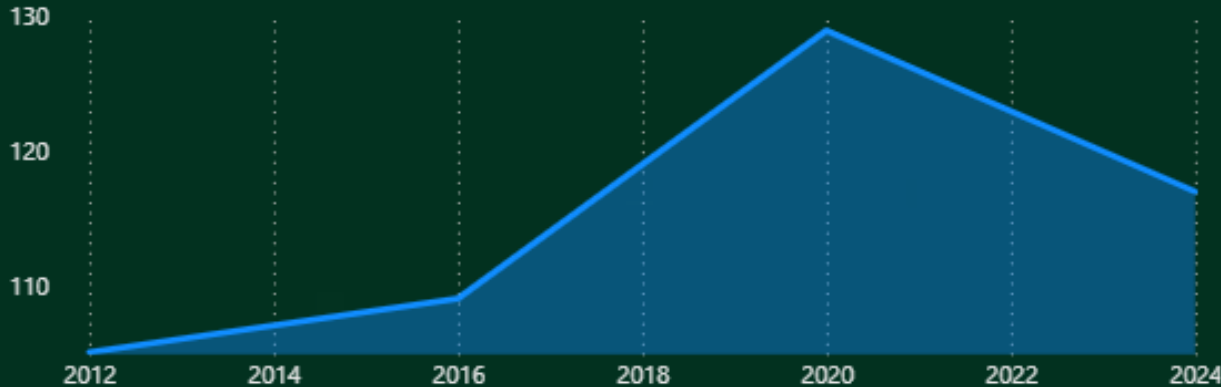
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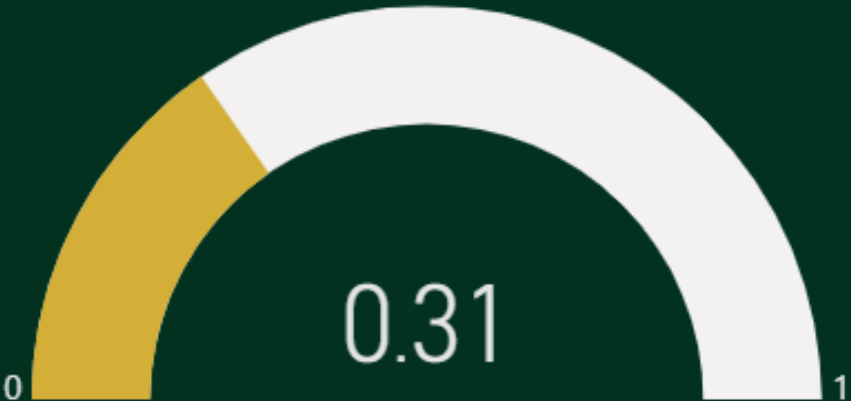
Female

Male

Medal Performance Trend Over Olympic Years



Overall Medal Achievement Rate



Question 5

What are the key factors associated with Olympic medal success?

Insights

- The overall medal achievement rate is **31%**, indicating that nearly one out of every three athletes secured a medal.
- Average medal-winning athletes have balanced physical characteristics, with an average height of approximately **175 cm**, weight of **72 kg**, and average age of **28 years**.
- Male and female athletes contributed almost equally to the total medal-winning population.
- Medal-winning performances increased steadily until the 2020 Olympic Games, followed by a slight decline in 2024.
- Physical attributes alone are insufficient predictors of Olympic success, highlighting the importance of training, experience, and sport specialization.

Key Insight

- Olympic success is driven by a combination of athlete preparation, sport expertise, and consistent performance rather than any single physical characteristic.